

Leader, HPC & AI, Lenovo ISG, Asia-Pacific: “It is not the technology that differentiates HPC and HPCaaS, but the way an organisation uses and funds it—which means it is either a capex or an opex strategy.”

As TVS’s use case demonstrates, HPC is not your usual network of desktop computers running ERP or CRM software. It is immensely powerful. This power is needed, say, for drug discovery of Covid-19, which requires running a few trillion calculations with accurate simulation and modelling in a short period of time. “HPC conducts complex calculations at high speeds across multiple servers, simultaneously. Collectively known as clusters, these groups of servers are composed of hundreds or even thousands of servers connected through a network,” explains Sayed Peerzade, EVP & Chief Cloud Officer, Yotta Infrastructure, a managed data centre service provider. Many times, HPC networks are actually networks of supercomputers. Globally, such systems are used in multiple high-compute areas such as semiconductor design, streaming a live sporting event, oil and gas simulations, genomics, etc. India, too, has been using HPC in weather and disaster forecasting, academic research, etc., for decades.

We know what you’re thinking—such systems are expensive. Installing HPC systems can cost anywhere between \$25,000 and \$1 billion, plus operational and maintenance costs. That’s costly and complex, both factors influencing companies’ tech departments to veer towards HPCaaS. “HPCaaS removes some of the complexities of designing and deploying an HPC data centre, as well as running optimised HPC applications,” explains Alex Norton, Research Director, Lead AI and Co-Lead Cloud Analyst at Hyperion Research. Available as a service through on-prem or in the cloud, on a monthly subscription model (eliminating upfront investment), it makes compute-intensive processing possible for enterprise and academia. In HPCaaS, all the HPC infrastructure including switching, routing, internet bandwidth, intrusion protection systems, etc., are managed and delivered by vendors as a part of the subscription. Scaling up, and incremental payment, is flexible based on an enterprise’s workload change.

According to Hyperion Research, the on-premise HPC server market was \$13.5 billion in 2020 globally, projected to grow to nearly \$20 billion by 2025. Include add-on storage, applications, and maintenance services, and the on-premises HPC market is expected to reach \$40 billion by 2025, from \$26.3 billion in 2020. On the other hand, HPCaaS worldwide is around \$4.5 billion today and is expected to double to \$9 billion by 2025. In India, on-premise HPC spend is around \$400 million a year, projected to reach \$600 million a year by 2026. As for HPCaaS in India, it is expected to grow to an estimated \$102 million by 2025 from \$47 million in 2020. “HPCaaS is growing in India, but slightly slower than the rest of the world,” says Norton.

## AS HPCaaS IS STILL NASCENT IN INDIA, PLAYERS ARE COMING UP WITH NEW WAYS TO BRING AWARENESS & INCREASE ADOPTION

### Industry Adoption

Earlier, supercomputers were largely used for nuclear test simulation, mapping the human genome or resurrecting dinosaurs in movies, and reserved for government and medical researchers, academics, and innovative movie-makers. But the rise of data-intensive technologies such as artificial intelligence (AI) and machine learning (ML), which require massively parallel compute (MPC) capabilities, has resulted in a wide range of organisations examining HPC solutions. These are being used for varied cases such as computer-aided design and engineering, autonomous driving, production optimisation, predictive maintenance, drug discovery, precision medicine, fraud and anomaly detection, treasury and trading analytics, Internet of Things (IoT), smart cities, and more.

While HPCaaS follows a cloud-based delivery model, these sophisticated systems feature an advanced combination of hardware as well as software. The key hardware for this includes high-performance CPUs, fabric, memory and storage, networking components, plus accelerators for specialised workloads. This high-performance infrastructure at the core is backed by data connectivity and redundant availability. HPC platform software, libraries, optimised frameworks for big data and deep learning, and other software tools, are value-adds one gets access to while signing up for HPCaaS.

What’s the outcome? Stunningly complex work can be done, relatively affordably, on the cloud. For example, Samsung Electronics is providing its fabless customers a virtual environment—Samsung Advanced Foundry Ecosystem (SAFE) Cloud Design Platform (CDP)—to design chips in the cloud in collaboration with Rescale. By adopting CDP, Samsung’s customers can reduce the burden of building their own server infrastructure,